

S02 MAXIMUM NOISE LEVELS | O (MAX : 3 PT)

Intent : Achieve desired ambient noise levels such that HVAC, exterior noise intrusion or other noise sources do not impact occupant health and well-being.

Summary : This WELL feature prescribes maximum thresholds for ambient background noise that correspond to optimal levels of interior and exterior noise exposure.

Issue : All spaces have some degree of ambient background noise from HVAC equipment, exterior sources (e.g., traffic, outdoor equipment, pedestrians) or other building services. When the sum of these noise sources exceeds comfortable levels, the space may not function as intended. For instance, elevated levels of background noise can impact the perception of public address systems and diminish the perception of spoken word, which reduces critical listening ability and task performance.¹⁻⁶ Studies indicate that employees are unable to habituate to noise in office environments over time and office noise, with or without speech, can create stress and disrupt performance on more complex cognitive tasks (e.g., memory of prose, mental arithmetic).⁷⁻¹¹ Studies have shown that exposure to traffic noise can lead to increased risk of cardiovascular system issues, diabetes, hypertension, stroke, depression and high blood pressure.¹²⁻¹⁷ For children, chronic aircraft noise exposure impairs reading comprehension, mental arithmetic and proofreading.^{3,18-21}

Solutions : Interior noise sources can be controlled by selecting HVAC equipment with lower sound ratings and by designing the system to reduce sound within ducts.²² Exterior noise can be controlled by providing sound reduction at the building façade, windows and any exterior penetrations.²³ In both cases, these sound sources are easier to control when considered at the earliest possible stages of design.²⁴⁻²⁶

PART 1 LIMIT BACKGROUND NOISE LEVELS (MAX : 3 PT)

For All Spaces except Dwelling Units & Guest Rooms:

Note :

Category 1 Room Types:

- Areas for conferencing
- Areas for learning
- Areas for speaking

Category 2 Room Types:

- Enclosed areas for concentration

Category 3 Room Types:

- Open areas for concentration
- Areas with regularly used PA systems
- Areas for dining (excluding kitchenettes)

Category 4 Room Types:

- Areas with machinery and appliances used by occupants (e.g., baggage handling areas, security, commercial kitchens, kitchenettes, labs where spoken lectures do not take place)

The following requirement is met:

- a. Background noise levels do not exceed the following thresholds, as applicable:

	Tier Sound Pressure Level (SPL)	Category 4	Category 3	Category 2	Category 1 Points:
1	Average SPL (Leq)	dBA 55	50	45	40
		dBC 75	70	65	60
	90th percentile SPL (L10)	dBA 65	60	55	50
	Average SPL (Leq)	dBA 50	45	40	35
2		dBC 70	65	60	55
	90th percentile SPL (L10)	dBA 60	55	50	45

For Dwelling Units & Guest Rooms:

The following requirement is met:

- a. Average background noise levels in bedrooms, when measured over a 12-hour minimum time period (which must include the hours of 10 pm to 7 am), do not exceed 35 dBA (Leq).²⁷

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Larsen JB, Vega A, Ribera JE. The Effect of Room Acoustics and Sound-Field Amplification on Word Recognition Performance in Young Adult Listeners in Suboptimal Listening Conditions. *Am J Audiol.* 2008;17(1):50-59.
2. Jennings MB. Universal Design for Hearing: Considerations for Examining Hearing Demands and Developing

Hearing Friendly Workplaces. AudiologyOnline. 2009.

3. Cheesman MF, Jennings MB, Klinger L. Assessing communication accessibility in the university classroom: Towards a goal of universal hearing accessibility. *Work*. 2013;46(2):139-150.
4. Way TJ, Long A, Weihing J, et al. Effect of noise on auditory processing in the operating room. *J Am Coll Surg*. 2013;216(5):933-938. doi:10.1016/j.jamcollsurg.2012.12.048
5. Weinstein ND. Effect of Noise on Intellectual Performance. *J Appl Psychol*. 1974;59(5):548-554. <http://search.ebscohost.com/login.aspx?direct=true&db=bth&AN=12424317&site=ehost-live&scope=site>.
6. Katz JD. Noise in the operating room. *Anesthesiology*. 2014;121(4):894-898. doi:10.1097/ALN.0000000000000319
7. Jahncke H, Hygge S, Halin N, Green AM, Dimberg K. Open-plan office noise: Cognitive performance and restoration. *J Environ Psychol*. 2011;31(4):373-382. doi:10.1016/j.jenvp.2011.07.002
8. Evans GW, Johnson D. Stress and open-office noise. *J Appl Psychol*. 2000;85(5):779-783. doi:10.1037/0021-9010.85.5.779
9. Pierrette M, Parizet E, Chevret P, Chatillon J. Noise effect on comfort in open-space offices: development of an assessment questionnaire. *Ergonomics*. 2015;58(1):96-106. doi:10.1080/00140139.2014.961972
10. Navai M, Veitch J a. Acoustic Satisfaction in Open-Plan Offices: Review and Recommendations. 2003:23. doi:<http://doi.org/10.4224/20386513>
11. Banbury S, Berry DC. Habituation and dishabituation to speech and office noise. *J Exp Psychol Appl*. 1997;3(3):181-195. doi:10.1037/1076-898X.3.3.181
12. Heidemann C, Niemann H, Paprott R, Du Y, Rathmann W, Scheidt-Nave C. Residential traffic and incidence of Type 2 diabetes: the German Health Interview and Examination Surveys. *Diabet Med*. 2014;31(10):1269-1276. doi:10.1111/dme.12480
13. Sørensen M, Hvidberg M, Andersen ZJ, et al. Road traffic noise and stroke: A prospective cohort study. *Eur Heart J*. 2011;32(6):737-744. doi:10.1093/eurheartj/ehq466
14. Niemann H, Bonnefoy X, Braubach M, et al. Noise-induced annoyance and morbidity results from the pan-European LARES study. *Noise Heal*. 2006;8(31):63. doi:10.4103/1463-1741.33537
15. Babisch W, Beule B, Schust M, Kersten N, Ising H. Traffic noise and risk of myocardial infarction. *Epidemiology*. 2005;16(1):33-40. doi:10.1097/01.ede.0000147104.84424.24
16. Jarup L, Babisch W, Houthuijs D, et al. Hypertension and exposure to noise near airports: The HYENA study. *Environ Health Perspect*. 2008;116(3):329-333. doi:10.1289/ehp.10775
17. Chang TY, Lai YA, Hsieh HH, Lai JS, Liu CS. Effects of environmental noise exposure on ambulatory blood pressure in young adults. *Environ Res*. 2009;109(7):900-905. doi:10.1016/j.envres.2009.05.008
18. Bronzaft AL, McCarthy DP. The effect of elevated train noise on reading ability. *Environ Behav*. 1975;7(4):517-528. doi:10.1177/001391657500700406
19. Hygge S, Evans GW, Bullinger M. A prospective study of some effects of aircraft noise on cognitive performance in schoolchildren. *Psychol Sci*. 2002;13(5):469-474. doi:10.1111/1467-9280.00483
20. HAINES MM, STANSFELD SA, JOB RFS, BERGLUND B, HEAD J. Chronic aircraft noise exposure, stress responses, mental health and cognitive performance in school children. *Psychol Med*. 2001;31(02):265-277. doi:10.1017/S0033291701003282
21. Belojević G, Öhrström E, Rylander R. Effects of noise on mental performance with regard to subjective noise sensitivity. *Int Arch Occup Environ Health*. 1992;64(4):293-301. doi:10.1007/BF00378288
22. Engineers TAS of HR and A-C, American Society of Heating Refrigerating and Air-Conditioning Engineers. Chapter 48: Noise and Vibration Control. In: *ASHRAE Handbook: HVAC Applications*. American Society of Heating Refrigerating and Air-Conditioning Engineers; 2015:1-28. <https://www.ashrae.org/technical-resources/ashrae-handbook/description-2015-ashrae-handbook-hvac-applications>.
23. The EP, Dep TNYC, York N. Noise Control for Building Exterior Heating , Ventilation and Air Conditioning Equipment Guidance Sheet. 2013:1-4.
24. Service GPB. Sound Matters. 2012;(January):1-42.
25. General Services Administration. Sound Matters: How to Achieve Acoustic Comfort in the Contemporary Office. 2011. https://www.wbdg.org/FFC/GSA/gsa_soundmatters.pdf.
26. Hodsmann P. Planning for Psychoacoustics: A Psychological Approach to Resolving Office Noise Distraction. 2015.

27. Joseph WS. Night noise guidelines for Europe. J Am Podiatr Med Assoc. 2009;100(5):1-162.
doi:10.1093/ejchocard/jer095